Response to Reviewers

IECCT 2026 — Paper ID: 1523

Progressive Token Pruning with Feature-Aware Knowledge Distillation for Efficient Vision Transformers

Dear Conference Chair and Reviewers,

We sincerely thank the reviewers for their thorough and constructive feedback. The comments have helped us substantially strengthen the paper. Below we address each reviewer comment individually, describe the specific revisions made, and indicate the locations of all changes in the revised manuscript.

All changes are highlighted in the revised manuscript for ease of review.

REVIEWER 1

Comment R1.1 — Significance of Contribution: Model Interpretability

> *More discussion on the impact of token pruning on model interpretability.*

Response: We thank the reviewer for raising this point. Progressive token pruning has a natural interpretability benefit: tokens that survive two rounds of L2-norm-based selection are precisely those representations that the network deems most semantically salient after deep contextual processing. This provides a post-hoc saliency map at no additional cost.

Revision: *Added a dedicated paragraph in Section V (Discussion) on interpretability: retained tokens after progressive pruning align with class-discriminative regions, offering a lightweight mechanism for spatial attention visualization without additional saliency modules. We also cross-reference Figure 5 (token visualization) as supporting evidence.*

Comment R1.2 — Significance of Contribution: Hyperparameter Tuning

> *Consider adding details on the hyperparameter tuning process.*

Response: We agree this was underspecified. The key hyperparameters are: pruning ratios (kappa_1=0.85, kappa_2=0.50), distillation temperature (T=4.0), feature loss weight (beta=0.3), and the alpha annealing schedule (0.70 to 0.50). These were selected via a small grid search over three candidate values each on a held-out 10% validation split of CIFAR-100 before final training.

Revision: *Added a new subsection IV-I 'Hyperparameter Selection' describing the grid search protocol, the candidate ranges explored, and the sensitivity of final accuracy to each parameter. Key finding: results are robust to moderate variation in T and beta, while the pruning ratios have stronger impact on both accuracy and FLOP reduction.*

Comment R1.3 — Significance of Contribution: Scalability

> *More analysis on the scalability of PTP-FAKD to larger models or datasets.*

Response: This is an important limitation we acknowledge. Our primary experiments use ViT-B/16 on CIFAR-100/10. The PTP framework is architecture-agnostic with respect to the number of encoder blocks (it partitions them into three equal stages), making it directly applicable to ViT-L/16 (24 blocks, 3 stages of 8) and ViT-H/14. The FLOP savings scale favorably with model size since attention cost grows as O(N^2 * d) — larger d amplifies the benefit of reducing N.

Revision: *Added a scalability analysis paragraph in Section V (Discussion) covering: (1) theoretical extension to ViT-L/16 and ViT-H/14, (2) expected FLOP reduction scaling, and (3) acknowledged that empirical validation on these models remains future work due to compute constraints.*

Comment R1.4 — Methodology: Robustness to Input Perturbations

> *How does PTP-FAKD affect the model's robustness to input perturbations?*

Response: This is a compelling direction. While we did not conduct formal adversarial robustness experiments, the implicit regularization effect of progressive token pruning — evidenced by our accuracy improvement over the dense baseline — suggests improved robustness. The student is forced to classify from semantically concentrated tokens, reducing reliance on background and texture cues that adversarial perturbations typically exploit. This is consistent with findings in MAE and sparse training literature.

Revision: *Added a discussion on robustness in Section V, connecting the token sparsity regularization hypothesis to adversarial robustness literature, and identifying formal robustness evaluation (e.g., CIFAR-100-C corruption benchmark) as a concrete future work direction.*

Comment R1.5 — Methodology: Adaptation to Other Architectures

> *Can the approach be adapted to other attention-based architectures?*

Response: Yes — PTP requires only that the model consist of stacked transformer encoder blocks with a CLS token, making it directly applicable to DeiT variants. For Swin Transformer, which uses shifted window attention without a global CLS token, adaptation would require using average-pooled patch representations for scoring rather than CLS alignment, and partitioning window-attention stages rather than individual blocks. Hybrid CNN-ViT architectures would require applying PTP only to the transformer portion.

Revision: *Added an architectural generalization discussion in Section V covering DeiT (direct compatibility), Swin Transformer (required modifications), and hybrid architectures.*

Comment R1.6 — Methodology: Structured Sparsity Understanding

> *How do the results contribute to understanding structured sparsity in ViTs?*

Response: Our controlled ablation provides a clean decomposition that is itself a contribution to understanding: progressive pruning without any distillation (88.98%) outperforms the dense teacher (87.03%), establishing that structured sparsity applied at the correct depth acts as an implicit regularizer — not merely a compression technique. The token visualization in Figure 5 provides spatial evidence for why: late-stage pruning selects semantically coherent, object-centric regions rather than texture-driven patches.

Revision: *Strengthened the Discussion section to explicitly frame this contribution to the structured sparsity literature, drawing connections to MAE and stochastic depth findings.*

Comment R1.7 — Practical Relevance: Applications and Deployment

> *Discuss potential applications and implications in efficient ViT deployment.*

Response: PTP-FAKD has clear deployment relevance: (1) edge inference on mobile/embedded hardware where quadratic attention is the primary bottleneck; (2) real-time video analysis requiring per-frame inference under latency constraints; (3) medical imaging applications where both accuracy and compute budget are constrained. The 73.5% attention FLOP reduction maps directly to reduced memory bandwidth and inference time in variable-length attention kernel implementations.

Revision: *Added a dedicated 'Deployment Implications' paragraph in Section V covering edge inference, video analysis, and medical imaging use cases, with a note on hardware prerequisites for realizing latency gains.*

Comment R1.8 — Practical Relevance: Comparison with Pruning Methods

> *Provide more details on the comparison with other pruning or compression methods.*

Response: Table III already contextualizes PTP-FAKD against DynamicViT, EViT, and SPViT at matched 50% token budgets. We acknowledge that direct comparison is complicated by differences in teacher accuracy and training recipes. We have expanded the table notes and added textual discussion clarifying these differences.

Revision: *Expanded Table III footnote and added a paragraph in Section IV-D explicitly discussing the methodological differences between our approach and each baseline, including parameter overhead (DynamicViT adds a learned predictor; PTP adds none), training complexity, and the specific conditions under which comparisons hold.*

Comment R1.9 — Practical Relevance: Real-World Datasets

> *Evaluate PTP-FAKD on real-world datasets or tasks (e.g., object detection).*

Response: We fully acknowledge this limitation. Extending to ImageNet-1K classification and COCO object detection would significantly strengthen the claims. However, these experiments require substantial compute (multi-GPU training for weeks) that was not available within our project constraints. We conducted our experiments on a single NVIDIA Tesla P100 (16GB) via Kaggle.

Revision: *Added explicit acknowledgment in the Limitations section that CIFAR experiments are a starting point, and that ImageNet-scale validation and downstream task evaluation (object detection) are the primary future work directions.*

REVIEWER 2

Comment R2.1 — Significance of Contribution: Explicit Novelty Statement

> *Please mention the novelty of the research work.*

Response: We thank the reviewer for this important point. The novelty of PTP-FAKD lies in two complementary contributions: (1) the insight that token pruning timing — not just budget — is the dominant determinant of accuracy, demonstrated through a clean 2x2 ablation; and (2) the targeted feature distillation at pruning boundaries specifically, rather than at arbitrary intermediate layers. Prior work either applies pruning at a single stage (DynamicViT, EViT) or uses distillation without tying supervision to the pruning schedule (SPViT). PTP-FAKD is the first to jointly optimize pruning schedule and distillation locality.

Revision: *Added a dedicated 'Summary of Novelty' paragraph at the end of Section I (Introduction), and a 'Distinction from Prior Work' paragraph in Section II-B explicitly contrasting PTP-FAKD against DynamicViT, EViT, and SPViT on the axes of (a) pruning stage, (b) scoring criterion, and (c) distillation design.*

Comment R2.2 — Methodology: Dataset Scale Limitation

> *Experiments are limited to CIFAR-10 and CIFAR-100, which are low-resolution, small-scale datasets. Recommendation: Evaluate on larger and more diverse datasets (ImageNet-1K, TinyImageNet, Flowers, etc.) to strengthen claims of generalization.*

Response: This is a valid and important limitation. CIFAR-100 was chosen because it enables controlled ablation experiments (6 configurations, each 25 epochs) within feasible compute budgets on a single P100 GPU. Full ImageNet-1K fine-tuning with comparable ablation depth would require approximately 10-20x more compute. We acknowledge that the generalization claim is currently supported only on CIFAR-scale data and that ImageNet validation is necessary to make stronger claims.

Revision: *Strengthened the Limitations section (Section V) with explicit acknowledgment of this constraint. Added a paragraph discussing expected behavior on ImageNet-1K: the progressive pruning benefit should persist since semantic representations in deeper ViT layers are more discriminative regardless of dataset scale, but the* *magnitude of improvement over the dense teacher may differ due to reduced overfitting pressure on larger datasets. TinyImageNet evaluation is identified as a tractable near-term extension.*

Comment R2.3 — Practical Relevance: Theoretical Basis for Implicit Regularization

> *The paper claims pruning acts as an implicit regularizer but does not provide theoretical insight (e.g., sparsity-induced smoothness, reduced overfitting, entropy analyses). Recommendation: Incorporate theoretical or empirical analysis supporting this interpretation.*

Response: We thank the reviewer for this substantive critique. We strengthen the theoretical grounding as follows: (1) Information-theoretic argument: by forcing predictions from a reduced token set, the model is trained to extract maximum mutual information between the CLS representation and the class label from fewer tokens — a form of information bottleneck regularization. (2) Empirical entropy analysis: we compute the entropy of the predicted class distribution for both the dense teacher and the PTP-FAKD student across the CIFAR-100 test set. The student produces lower-entropy (more confident) predictions on correct classifications, consistent with reduced reliance on spurious low-level cues. (3) Connection to dropout/stochastic depth: progressive token removal during training is structurally analogous to stochastic depth, which has established regularization benefits [Huang et al., 2016].

Revision: *Added a 'Theoretical Analysis' subsection in Section V providing: (a) the information bottleneck framing, (b) a table reporting mean prediction entropy for the dense teacher vs. PTP-FAKD student on the CIFAR-100 test set, and (c) formal connection to stochastic depth regularization with appropriate citation.*

SUMMARY OF ALL REVISIONS

The following table lists all changes made to the revised manuscript:

Comment	Location in Paper	Change Made
R1.1	Section V (Discussion)	Added interpretability paragraph; cross-ref to Fig. 5
R1.2	New Section IV-I	Added hyperparameter selection subsection with grid search details
R1.3	Section V (Discussion)	Added scalability analysis for ViT-L/16 and ViT-H/14
R1.4	Section V (Discussion)	Added robustness discussion; CIFAR-100-C as future work
R1.5	Section V (Discussion)	Added architectural generalization: DeiT, Swin, hybrid
R1.6	Section V (Discussion)	Strengthened sparsity contribution framing
R1.7	Section V (Discussion)	Added deployment implications paragraph
R1.8	Section IV-D + Table III	Expanded comparison discussion and table footnote
R1.9	Section V (Limitations)	Added real-world dataset limitation acknowledgment
R2.1	Section I + Section II-B	Added novelty summary and distinction from prior work
R2.2	Section V (Limitations)	Strengthened dataset scale limitation discussion
R2.3	New Section V subsection	Added theoretical analysis: info bottleneck, entropy table, stochastic depth connection

We believe the revised manuscript substantially addresses all reviewer concerns. We thank the Program Committee for the opportunity to improve the work and look forward to presenting at IECCT 2026.

Sincerely,

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